MEGA BIOINFORMATICS INTERNSHIP REPORT

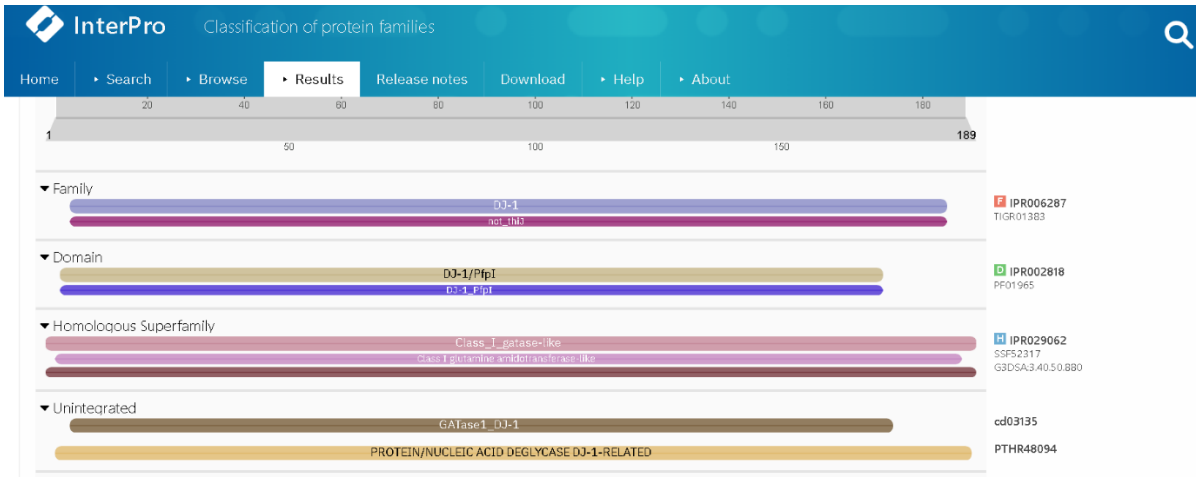
Submitted By,

NASRI THAHA

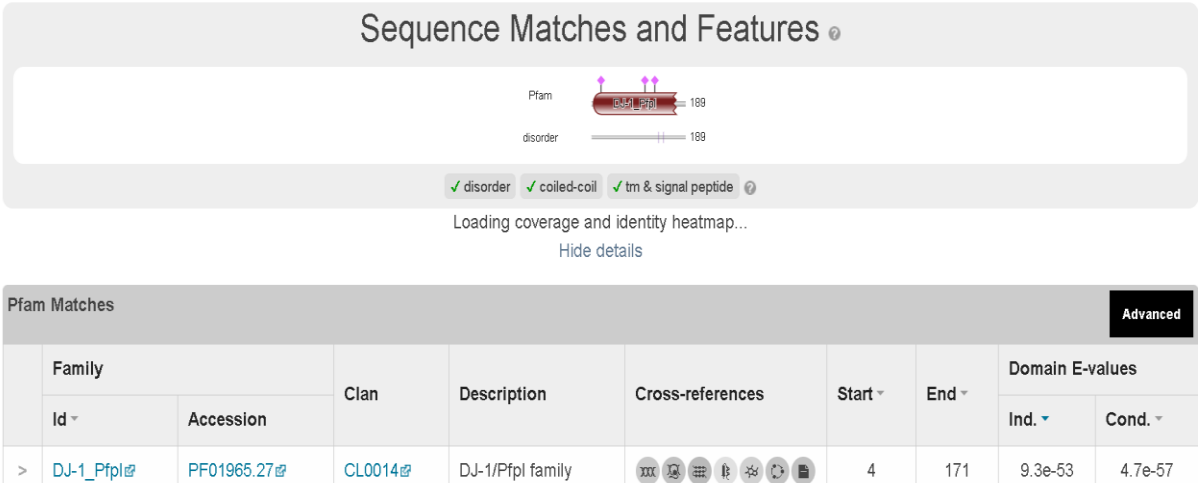
Day 2:

Gene Annotation

Protein Name: Parkinson disease protein 7 [Homo sapiens]
Protein ID - NP_009193
Find the following Gene function details
Location- Start = 4 and end 171
Family, peptidase C56 family
Protein/nucleic acid deglycase DJ-1 (IPR006287)



Clan, CL0014
Domain, DJ-1/PfpI family
E value 9.3e-53



	Family		Clan	Description	Cross-references	Start	End	Domain E-values	
	Id	Accession						Ind.	Cond.
>	DJ-1_PfpI	PF01965.27	CL0014	DJ-1/PfpI family		4	171	9.3e-53	4.7e-57

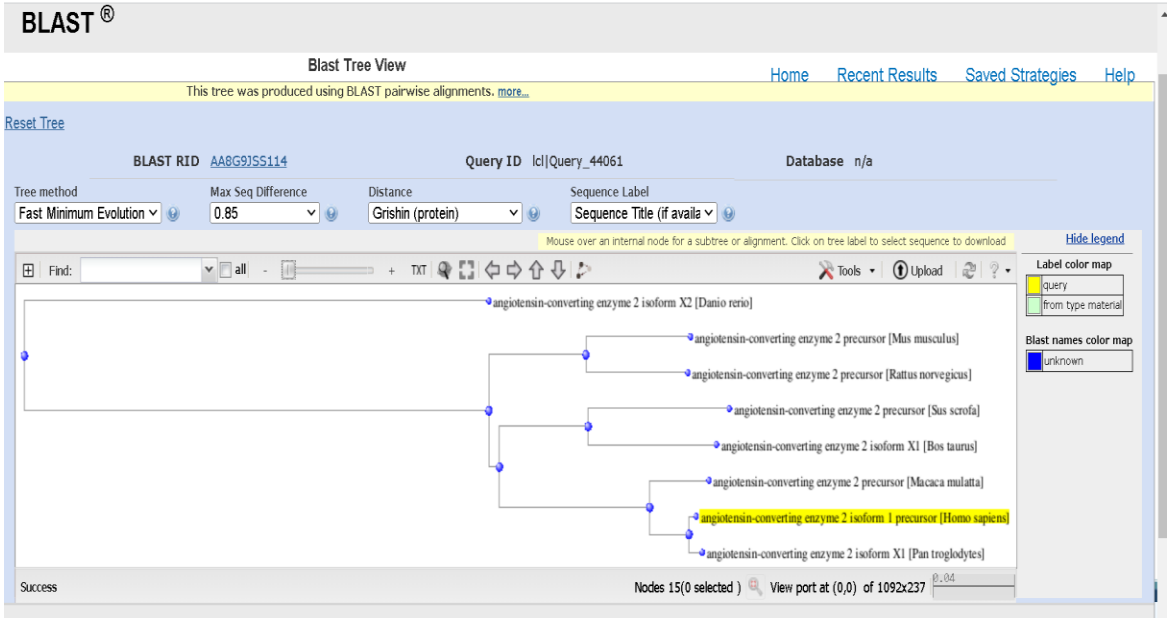
Description of function: Multifunctional protein with controversial molecular function which plays an important role in cell protection against oxidative stress and cell death acting as oxidative stress sensor and redox-sensitive chaperone and protease s a protein deglycase, repairs methylglyoxal- and glyoxal-glycated proteins, and releases repaired proteins and

lactate or glycolate, respectively. Deglycates cysteine, arginine and lysine residues in proteins, and thus reactivates these proteins by reversing glycation by glyoxals. Acts on early glycation intermediates (hemithioacetals and aminocarbinols), preventing the formation of advanced glycation endproducts (AGE) that cause irreversible damage lso functions as a nucleotide deglycase able to repair glycated guanine in the free nucleotide pool (GTP, GDP, GMP, dGTP) and in DNA and RNA. Is thus involved in a major nucleotide repair system named guanine glycation repair

Day 3: Phylogenetics

Construct a Phylogenetic tree for components of Corona virus.

You can choose any gene/protein/component associated with Corona virus for atleast 5 different species/variants. Add the screenshot of the tree here.



Day 4:

Genome name(the one of your interest): ACE2 - angiotensin converting enzyme 2
From RAST results:
Mention the desired nucleotide sequence that you choose to perform BLAST on and fill in the following:

1)Nucleotide sequence :
atgtttattttcttattatttcttactctcactagtggtagtgaccttgaccggtgcaccacttttgatgatgttcaagctcctaattacactcaac
atacttcatctatgagggggtttactatcctgatgaaatttttagatcagacactctttatttaactcaggatttatttcttccattttattctaatg
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tgactcattcaaagaagagctggacaagtacttcaaaaatcatacatcaccagatgttgatcttggcgacatttcaggcattaacgcttct
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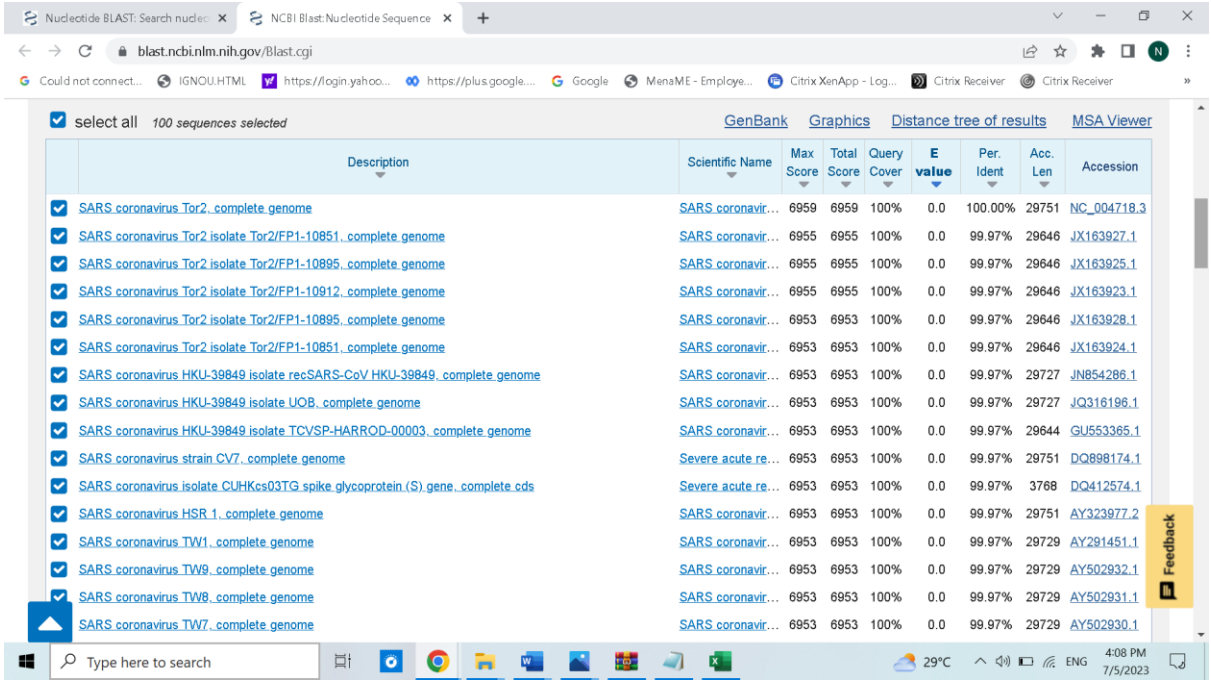
atgagcaatatattaaatggccttggtatgtttggctcggcttcattgctggactaattgccatcgtcatggttacaatcttgctttgttgcatg
actagttgttgcagttgcctcaagggtgcatgctcttgtggttcttgctgcaagtttgatgaggatgactctgagccagttctcaagggtgt
caaattacattacacataa

2)Location on the genome AY274119.3_21492_25259
3)Start of nucleotide -21492 End of nucleotide - 25259
4)Function - hypothetical protein

Perform a BLAST on the nucleotide sequence and paste a screenshot of the obtained BLAST results.



select all 100 sequences selected

GenBank Graphics Distance tree of results MSA Viewer

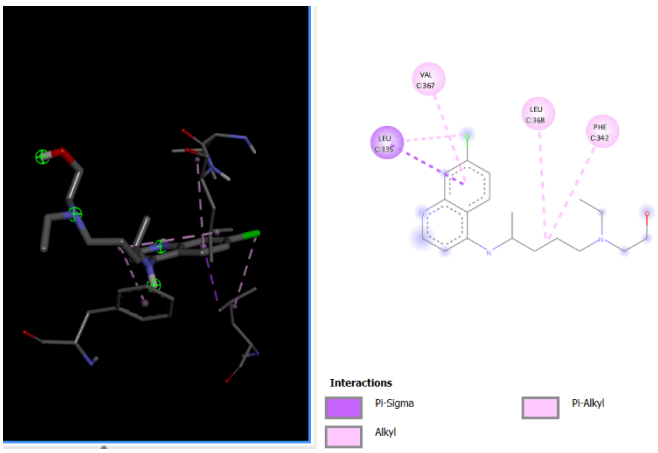
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
SARS coronavirus Tor2, complete genome	SARS coronavir...	6959	6959	100%	0.0	100.00%	29751	NC_004718.3
SARS coronavirus Tor2 isolate Tor2/FP1-10851, complete genome	SARS coronavir...	6955	6955	100%	0.0	99.97%	29646	JX163927.1
SARS coronavirus Tor2 isolate Tor2/FP1-10895, complete genome	SARS coronavir...	6955	6955	100%	0.0	99.97%	29646	JX163925.1
SARS coronavirus Tor2 isolate Tor2/FP1-10912, complete genome	SARS coronavir...	6955	6955	100%	0.0	99.97%	29646	JX163923.1
SARS coronavirus Tor2 isolate Tor2/FP1-10895, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29646	JX163928.1
SARS coronavirus Tor2 isolate Tor2/FP1-10851, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29646	JX163924.1
SARS coronavirus HKU-39849 isolate recSARS-CoV HKU-39849, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29727	JN854286.1
SARS coronavirus HKU-39849 isolate UOB, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29727	JQ316196.1
SARS coronavirus HKU-39849 isolate TCVSP-HARROD-00003, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29644	GU553365.1
SARS coronavirus strain CV7, complete genome	Severe acute re...	6953	6953	100%	0.0	99.97%	29751	DQ898174.1
SARS coronavirus isolate CUHKcs03TG spike glycoprotein (S) gene, complete cds	Severe acute re...	6953	6953	100%	0.0	99.97%	3768	DQ412574.1
SARS coronavirus HSR 1, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29751	AY323977.2
SARS coronavirus TW1, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29729	AY291451.1
SARS coronavirus TW9, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29729	AY502932.1
SARS coronavirus TW8, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29729	AY502931.1
SARS coronavirus TW7, complete genome	SARS coronavir...	6953	6953	100%	0.0	99.97%	29729	AY502930.1

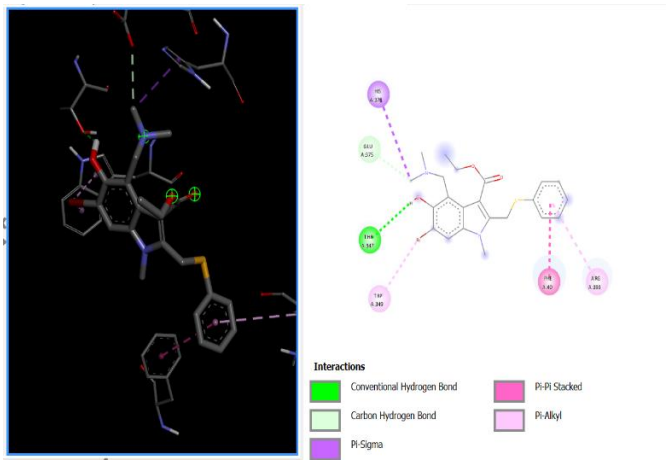
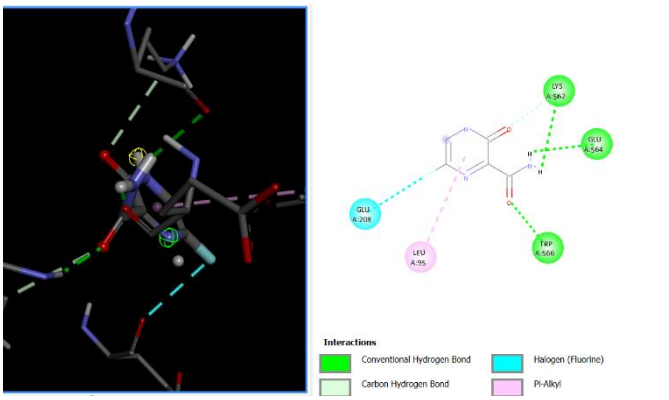
Day 5 & 6:

Molecular Docking

Protein Name: Angiotensin-converting Enzyme 2

Protein ID – 7dqa

Ligand Name	Ligand ID	Follows Lipinski Rule?	Energy value	Dock Image
hydroxychloroquine	3652	Yes; 0 violation	-6.1	

Arbidol	131411	Yes; 0 violation	-6.3	
Favipiravir	492405	Yes; 0 violation	-5.2	

Day 7: Heatmap analysis
Objective : To plot a heat map and understand the differential expression based on numbered data.

Problem statement Or research question: To identify over expression of genes in different strains.

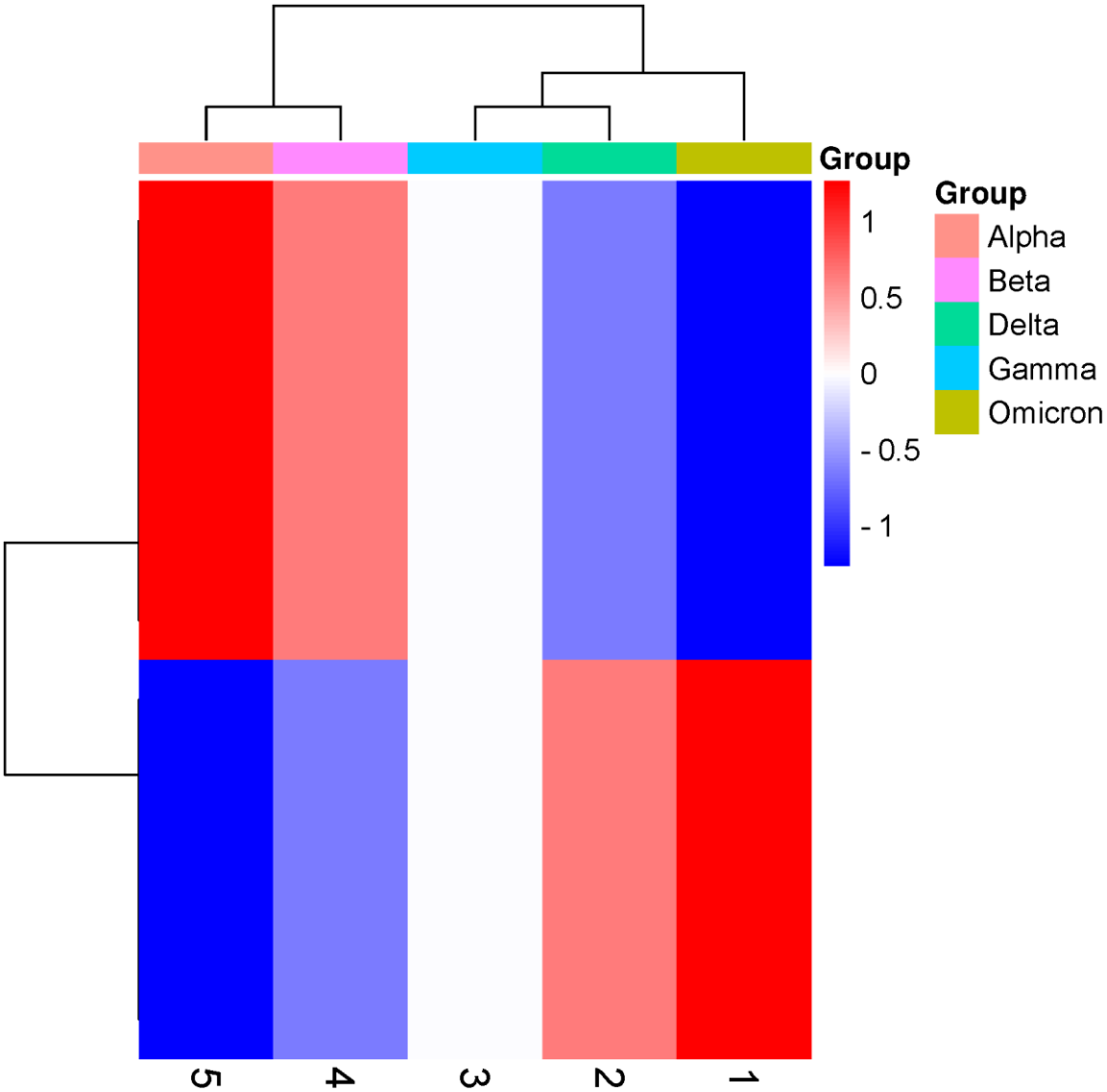
Input details
1. Gene(s) Name: ACE 2
2. Protein(s) Name (transcriptome details): ACE 2

Input data Table

ORF	Alpha	Beta	Gamma	Delta	Omicron
ORF1a	5	4	3	2	1
ORF1b	4	3	2	1	0
ORF3a	3	2	1	0	-1
ORF6	2	1	0	-1	-2

ORF7a	1	0	-1	-2	-3
ORF7b	0	-1	-2	-3	-4
ORF8	-1	-2	-3	-4	-5
N	0	1	2	3	4
M	1	2	3	4	5
S	2	3	4	5	6
L	3	4	5	6	7
ORF10	4	5	6	7	8

Output heatmap:

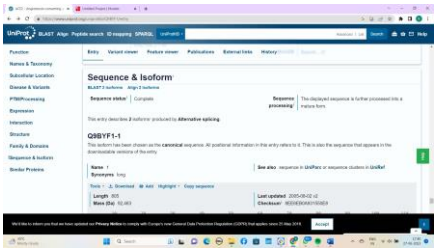
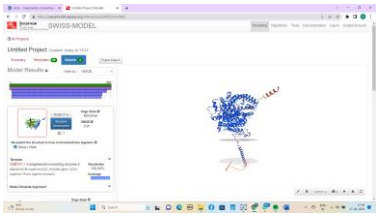


Discussion points: To study GMO, to study epigenetic effects and to know how are biomarkers found.

Five interpretation points understood:
1) Alpha has stronger expression.
2) ORF1a is most virulent in Alpha than ORF8 in Omicron.
3) ORFN to ORF10 are stronger ORFs in these strains.
4) ORF1a to ORF8 has very less impact though they are present in all other strains.
5) ORF8 is more expressed in Alpha strain than in Omicron.

Day 8 & 9: Homology Modelling:
You can choose any protein which is involved in SARS CoV-2 Pathogenesis (Eg: ACE2 receptor, Any envelope protein) and can take at least 2 homologous sequences with sequence similarity >30%. Try to develop an hypothesis around it (Like Why you want to use Homology modelling for your protein of interest, Purpose and outcome of it) and more importantly how it is going to add value to your hypothesis.

Protein: Angiotensin-converting Enzyme 2
Gene: ACE2
PDB: Q9BYF1 (First Isoform)

Target	Sequence	Result
		

Day 10:

Please paste your GitHub account link:

https://github.com/NasriThaha